Modelling the Variance: ARCH / GARCH Models

These notes accompany Lecture 11. They combine the slide material with the explanations given in class. Terms printed in bold are defined in the glossary at the end.

Every model so far has treated the variance of the disturbance as a nuisance – something to be corrected for, as in Lecture 10. Here the variance becomes the object of interest in its own right. Financial econometrics needs models that describe investors' attitude not only to expected returns but to risk, and that is what the ARCH family provides.

Learning outcomes – after this lecture you should be able to:

1. Explain volatility clustering and distinguish conditional from unconditional variance.
2. Write down ARCH(q) and GARCH(p, q) models and state the parameter restrictions each requires.
3. Carry out the ARCH LM test and interpret its result.
4. Interpret estimated ARCH and GARCH coefficients, and compute persistence.
5. Diagnose non-normal residuals and choose between robust standard errors and a Student-t specification.
6. Explain the leverage effect, test for it, and interpret an EGARCH model.

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1. Volatility clustering

Conventional econometric analysis views the variance of the disturbance as constant over time – the homoskedasticity assumption. But financial and many economic time series show periods of unusually high volatility followed by calmer periods of low volatility.

A quick look at daily FTSE-100 returns makes it obvious: certain stretches are visibly riskier than others, and those riskier times are *concentrated*, followed by concentrated periods of lower volatility. This is what financial analysts call **volatility clustering**.

1.1 Conditional versus unconditional variance

> **The distinction the whole lecture rests on**
>
> Over long periods the variance can still be treated as constant – that is the unconditional variance, the long-run forecast.
>
> Over short periods it changes. That is the conditional variance, based on our best model of the series and on what has just happened.
>
> Homoskedasticity is a statement about the first. Volatility clustering is a statement about the second. They are entirely compatible.

Consider an investor planning to buy an asset at time t and sell it at t + 1. A forecast of the expected return alone is not enough – she needs to know the variance of the return over the holding period. The unconditional variance is of no use for that; what matters is the conditional variance at that particular moment.

1.2 Why forecast the variance?

- **To measure risk.** The variance of price changes is a measure of the risk of owning an asset. An investor worried about risk would be less willing to participate during a period of high rather than low volatility.
- **To price derivatives.** The value of options depends on the variance of the underlying asset, so an options trader wants the best available forecast of future volatility.
- **To get forecast intervals right.** If the variance of the forecast error changes over time, the width of the interval should change too – wide during turbulent periods, tighter during calm ones.

2. The ARCH model

2.1 ARCH(1)

Engle (1982) proposed modelling the mean and the variance of a series simultaneously when the conditional variance is not constant. The model has two equations:

$$\textit{Mean:} \quad y_t = \beta_0 + \beta_1 y_{t-1} + \beta_2 x_t + u_t, \quad u_t \mid \Omega_t \sim iid\ N(0, \sigma_t^2)$$

$$\textit{Variance:} \quad \sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2$$

where Ω_t is the information set. The variance of the residual at time t depends on the squared error from the previous period: when a big shock happens at t − 1, the value of u_t is more likely to be large in absolute terms too.

Restriction	Why
$\alpha_0 > 0$	Sets the floor for the variance – a variance cannot be negative by definition
$0 \le \alpha_1 < 1$	Non-negativity keeps the variance positive; below one ensures the errors are weakly stationary
$\lvert\beta_1\rvert < 1$	Ensures y_t itself is stationary, conditional on x_t being stationary

Under these conditions the unconditional variance is finite:

$$Var(u_t) = \alpha_0 / (1 - \alpha_1)$$

2.2 ARCH(q)

The conditional variance can depend on more than one lagged squared error. In general:

$$\sigma_t^2 = \alpha_0 + \Sigma_{j=1..q}\, \alpha_j u_{t-j}^2$$

with $\alpha_0 > 0$ and $\alpha_j \ge 0$ for all j. The covariance-stationarity condition is now on the sum:

$$\Sigma_{j=1..q}\, \alpha_j < 1, \quad \text{so that } Var(u_t) = \alpha_0 / (1 - \Sigma\alpha_j)$$

If the sum exceeds one, the unconditional variance of the residuals is infinite.

2.3 Testing for ARCH effects

Before estimating an ARCH model it is important to check whether ARCH effects are present at all – that is what tells you an ARCH estimation method is needed rather than plain OLS.

1. Estimate the mean equation as usual – construct the best regression model for y – and obtain the residuals.
2. Run the auxiliary regression of the squared residuals on their own lags and a constant: $\hat{u}_t^2 = \delta_0 + \delta_1 \hat{u}_{t-1}^2 + \ldots + \delta_q\, \hat{u}_{t-}q^2 + v$.
3. Form the hypotheses. H_0: $\delta_1 = \delta_2 = \ldots = \delta_q = 0$ (no ARCH effects – residuals are homoskedastic) against H_1: at least one $\delta \neq 0$.
4. Compute $LM = n \cdot R^2$ from the auxiliary regression. Under H_0 it follows a χ^2 distribution with q degrees of freedom.
5. Reject H_0 if the p-value falls below the chosen significance level.

2.4 Worked example: Apple stock

The data are daily Apple closing prices in dollars. The price has an upward trend and is non-stationary; the first difference is essentially stationary, so we work with daily returns (100*log-diff). The time plot of returns shows clear stretches of larger and smaller volatility, so ARCH effects look likely.

The mean equation here is an MA(1). Its results are not of interest in themselves – what matters is whether ARCH effects remain in its residuals.

Which lag length to test

Test at lag 1 as the baseline, since first-order ARCH is the canonical case. Volatility clustering in daily returns is usually strongly detectable at the first lag – if an ARCH effect is there, lag 1 will almost always catch it.

Then test at a longer horizon matching the data frequency, to check for residual structure the model missed. For daily data q = 5 (one trading week) is standard; q = 10 and q = 22 (about two weeks and one trading month) are also commonly reported.

At lag 1 the p-value is 0.000, so we reject the null of no ARCH effects and estimate an ARCH(1). In EViews: click Estimate to return to the Equation Specification window and change the method to ARCH; the upper part specifies the mean equation and the lower part the variance equation. For a simple ARCH(1) set the ARCH order to 1 and GARCH to 0.

The estimate of α_1 is highly significant, positive and less than one; the constant is positive too. Re-testing the standardized residuals at lag 1 now fails to reject, so the ARCH(1) has absorbed the first-order effect. But testing at lag 5 rejects again (p = 0.000) – higher-order effects remain, and an ARCH(5) is needed.

Reading the conditional volatility series

The fitted conditional standard deviation σ_t is the model's day-by-day estimate of risk, in the same units as the return. An average value of 1.7 means the model expects a typical daily move of about ±1.7%; at the peaks it braces for swings near ±7%.

What the picture shows

Risk is not constant. Unlike the fixed variance assumed in ordinary regression, volatility moves through time – and it moves in a distinctive way: high-volatility days arrive in clusters rather than at random.

The line rises sharply but falls slowly. Shocks push volatility up abruptly, then it decays gradually back toward its long-run average of about. That slow decay is what persistence looks like.

The largest spike, in early 2020, is the COVID shock. The model knows nothing about pandemics – yet by reacting only to the size of recent returns it independently reconstructs the market's major stress episodes. This is why conditional volatility is the natural input to risk measures such as Value-at-Risk.

2.5 The drawback of ARCH

Engle himself noted that the ARCH specification looks more like a moving average than an autoregression. The practical problem follows from that: volatility in daily returns is highly persistent, with memory stretching over dozens of days, and a finite sum of a few squared-error terms cannot reproduce that slowly decaying dependence. A pure ARCH(q) needs too many lags.

The solution, worked out by Tim Bollerslev in 1986, was to include lagged conditional variance terms as autoregressive terms – letting today's variance depend on yesterday's variance.

3. The GARCH model

3.1 GARCH(1,1)

$$\sigma_t^2 = \alpha_0 + \varphi_1 \sigma_{t-1}^2 + \alpha_1 u_{t-1}^2$$

The conditional variance is now a function of three terms:

Term	Name	What it captures
α_0	Constant	The floor keeping the variance bounded away from zero
$\alpha_1 u_{t-1}^2$	ARCH term	News about volatility from the previous period – the size of yesterday's shock
$\phi_1 \sigma_{t-1}^2$	GARCH term	Last period's forecast variance – the inertia of volatility

The restrictions are $\alpha_0 > 0$, $\alpha_1 \geq 0$ and $\phi_1 \geq 0$ to guarantee a positive variance for any shock, plus the covariance-stationarity condition:

$$\alpha_1 + \varphi_1 < 1, \quad \text{giving} \quad Var(u_t) = \alpha_0 / (1 - \alpha_1 - \varphi_1)$$

This is what makes the unconditional variance finite and positive, and gives mean-reverting rather than explosive volatility.

> **Why GARCH(1,1) is so widely used**
>
> The GARCH(1,1) specification is equivalent to an infinite-order ARCH model with coefficients declining geometrically.
>
> So it captures the long, slowly-fading memory of volatility with two parameters instead of dozens – fewer parameters to estimate, and fewer degrees of freedom lost.

The general GARCH(p, q) allows p autoregressive GARCH terms and q moving-average ARCH terms, with the stationarity condition $\Sigma\phi_i + \Sigma\alpha_j < 1$.

3.2 Worked example: Apple daily returns

The mean equation here is trivial – just a constant, with $R^2 \approx 0$. That is entirely normal: daily returns are nearly unpredictable in the mean. All the interesting structure is in the variance.

Parameter	Estimate	Reading
Constant α_0	0.14235	Positive, as required – the floor for the variance
ARCH term α_1	0.09873	About 10% of yesterday's squared surprise carries into today's variance
GARCH term ϕ_1	0.85624	Persistence – 86% of yesterday's variance carries forward
$\alpha_1 + \phi_1$	0.95497	Below one, so stationary and mean-reverting – but high, so shocks die out only slowly

All three coefficients are strongly significant. The dominance of φ_1 over α_1 (0.86 against 0.10) tells the story: volatility is driven far more by its own inertia than by fresh daily shocks. It is smooth and slow-moving, not jumpy.

3.3 Evaluating the model

Residual diagnostics

The correlogram of standardized residuals checks whether any linear dependence remains in the mean. The Q-statistic probabilities are all large, so no significant autocorrelation is left and the mean equation is adequate. The ARCH LM test on the standardized residuals shows no remaining conditional heteroskedasticity – the GARCH(1,1) has absorbed the volatility clustering.

Non-normality – and two different fixes

The residuals are not normally distributed. This does not break the coefficient estimates: under quasi-maximum-likelihood the estimator remains consistent and asymptotically normal even when the assumed error distribution is wrong. What it does break is the inference on α and ϕ.

Fix	What it repairs	What it does not repair
Robust standard errors	Inference on the parameters. Coefficients unchanged, standard errors somewhat larger	The assumed distribution stays normal, so tail probabilities remain wrong
Student-t errors	The assumed distribution itself – heavier tails assign realistic probabilities to large shocks	–

> **Which one you need depends on your question**
>
> If your use case is describing volatility dynamics and testing whether ARCH and GARCH effects matter, robust standard errors are enough and you are done.
>
> If you are going to compute Value-at-Risk, they are not. Daily stock returns have fat tails – extreme moves happen far more often than a normal distribution predicts – and robust errors leave the assumed distribution normal. VaR is a statement about the tail, so the tail has to be right.

Re-estimating with Student-t errors, the key new parameter is the degrees of freedom, estimated at about 4.2 and highly significant. A high ν means close to normal – at $\nu \approx 30$ the t is practically indistinguishable from the normal – so $\nu \approx 4$ is strong evidence of fat tails. Both AIC and Schwarz fall sharply, so the t-distribution is clearly preferred. Notice that the volatility dynamics barely change: the ARCH and GARCH coefficients, and the high persistence, are almost the same as before.

3.4 ARCH(5) versus GARCH(1,1)

Plotting the two fitted volatility series side by side shows why GARCH won.

	ARCH(5)	GARCH(1,1)
Shape	Spiky – jumps up and collapses back quickly	Smooth – rises then decays gradually into rounded humps
Why	Computes today's variance from only the last five squared shocks: a five-day memory that forgets each shock as it rolls out of the window	The recursive term makes today's variance depend on yesterday's variance, giving a long, slowly-fading memory
Verdict	Abrupt spikes are the visual symptom of a memory too short to hold the persistence	The drawn-out decay reflects the real clustering in the data

Both models identify the same turbulent episodes – they agree on when risk was high. They disagree about how long it lasted, and the data side with GARCH.

4. Asymmetry and the EGARCH model

4.1 The problem: GARCH treats good and bad news alike

A major restriction of the ARCH and GARCH specifications is that they are symmetric. The shock enters squared, so the model sees only its magnitude, never its sign. A +5% day and a −5% day have exactly the same effect on tomorrow's volatility.

But for equities it has long been observed that negative shocks – bad news – have a larger impact on volatility than positive shocks of the same magnitude. A symmetric GARCH simply cannot represent this.

4.2 The leverage effect

The classic explanation (Black, 1976) runs through the balance sheet:

$$\textit{Price falls} \rightarrow \textit{debt-to-equity ratio rises} \rightarrow \textit{equity gets riskier} \rightarrow \textit{volatility rises}$$

Good news does the reverse but more weakly. More broadly this is a volatility asymmetry, driven by risk premia, panic selling and market psychology as much as by accounting leverage – but the label "leverage effect" has stuck regardless of mechanism. Empirically it is one of the most robust stylised facts of equity returns.

4.3 Testing for asymmetry: the Engle-Ng sign-bias test

Before adopting a more complex model, test formally whether asymmetry is present. Starting from a GARCH(1,1) fitted to daily FTSE-100 returns, the test regresses squared residuals on the sign and size of the lagged shock:

$$\hat{u}_t^2 = c + b_1 S^-_{t-1} + b_2 S^-_{t-1}\varepsilon_{t-1} + b_3 S^+_{t-1}\varepsilon_{t-1} + \textit{error}$$

where $S^-_{t-1} = 1$ when $\varepsilon_{t-1} < 0$ and zero otherwise, and $S^+_{t-1} = 1 - S^-_{t-1}$.

> **Why only one level dummy appears**
>
> S^+ and S^- are perfectly complementary: for every observation exactly one is 1, so $S^+ + S^- = 1$ always. Including c, S^- and S^+ together would produce perfect multicollinearity.
>
> One level dummy is therefore dropped and its role absorbed into the constant. So the constant measures the average squared residual following a positive shock, and the S^- coefficient measures how much that average differs following a negative shock.

The result: the effect lives in the negative-size bias – larger price drops drive volatility up, while positive shocks of the same size do not. The joint test rejects symmetry decisively ($p < 0.001$). A symmetric GARCH is misspecified for this series.

4.4 Exponential GARCH

$$\log(\sigma_t^2) = \alpha_0 + \alpha_1 |z_{t-1}| + \gamma z_{t-1} + \varphi_1 \log(\sigma_{t-1}^2)$$

where $z_{t-1} = u_{t-1} / \sigma_{t-1}$ is the standardized shock. Two features do the work:

- **The sign term.** The extra term $\gamma \cdot z_{t-1}$ carries the sign of the shock, so good and bad news can differ. If $\gamma = 0$ the model is symmetric; a negative γ means bad news raises volatility more.
- **The log variance.** Modelling $\log(\sigma^2)$ guarantees $\sigma^2 = \exp(\cdot) > 0$ automatically, so no non-negativity constraints are needed and the coefficients may take any sign.

Note that α_1 multiplies the absolute value $|z_{t-1}|$ – this is the size or magnitude effect, and a positive α_1 is what you expect, since bigger surprises mean more volatility. The signed term next to it is what creates the asymmetry. Without the absolute value the two terms would collapse into one and there would be no asymmetry at all.

In this log formulation the covariance-stationarity condition is simply $\phi_1 < 1$.

4.5 EGARCH for the FTSE-100

A negative intercept is not a problem here, because the model works in logs – it pins down the long-run level of log-variance. The two coefficients of interest are the size effect and the asymmetry term, both strongly significant, with the asymmetry term negative as the leverage effect predicts.

Shock	Calculation	Impact on log-variance
Positive (+)	$\alpha_1 + \gamma = 0.1725 + (-0.1428)$	0.030
Negative (−)	$\alpha_1 - \gamma = 0.1725 - (-0.1428)$	0.315

So a price drop raises FTSE volatility roughly ten times as much as an equal-sized rise. That ratio is the leverage effect measured, not merely asserted.

5. Alternative specifications

- **GARCH-M** – allows the conditional mean to depend on its own conditional variance. Risk-averse investors require a premium to hold a risky asset, and that premium is a positive function of the risk. If risk is captured by the conditional variance, the variance belongs in the mean equation.
- **Explanatory variables in the mean equation** – for example, uncovered interest parity terms. ARCH and GARCH models can be quite sensitive to how the mean equation is specified.
- **Explanatory variables in the variance equation** – global shocks, for instance, may affect the volatility of a national stock index.

6. Summary: the estimation procedure

Stage 1 – identifying conditional heteroskedasticity

1. Construct the best model for the series, in the ARIMA class or the regression class, and calculate its residuals.

2. Perform the ARCH LM test to detect conditional heteroskedasticity.

Stage 2 – estimating the ARCH/GARCH model

1. Estimate the model, checking that the intercept in the variance equation is positive and the ARCH and GARCH coefficients non-negative.
2. Check covariance-stationarity of both the mean and the variance equations.
3. Evaluate adequacy with the standard tools – correlogram of standardized residuals, ARCH LM test on standardized residuals, normality. Adjust the model if necessary.
4. Test for leverage effects, and estimate an EGARCH if they are present.

> **The through-line**
>
> ARCH said: today's variance depends on the size of recent shocks.
>
> GARCH added: and on its own past, which is what makes volatility persistent without needing dozens of lags.
>
> EGARCH added: and on the sign of the shock, because markets fear falls more than they celebrate rises.
>
> Each step was a response to something the previous model visibly failed to reproduce in the data. That is how model classes actually develop.

Key terms

Term	Meaning
Volatility clustering	Periods of high volatility followed by more high volatility, and calm followed by calm.
Conditional variance	σ_t^2, the one-period-ahead forecast variance based on past information.
Unconditional variance	The long-run variance, $\alpha_o/(1 - \Sigma\alpha - \Sigma\phi)$; can be constant even when the conditional variance moves.
ARCH(q)	Conditional variance depends on q lagged squared errors.
GARCH(p, q)	Conditional variance depends on its own p lags as well as q lagged squared errors.
ARCH term	The coefficient on the lagged squared residual; how strongly a fresh shock feeds into volatility.
GARCH term	The coefficient on the lagged conditional variance; the inertia of volatility.
Persistence	$\alpha_1 + \phi_1$ in a GARCH(1,1). Must be below one for stationarity; close to one means slow decay.
Covariance-stationarity condition	The restriction making the unconditional variance finite and positive.
ARCH LM test	Regress squared residuals on their own lags; LM = $n \cdot R^2 \sim \chi^2_q$. Null: no ARCH effects.
Standardized residual	$z_t = u_t / \sigma_t$ – the shock expressed in units of its own conditional standard deviation.

Term	Meaning
Quasi-maximum likelihood	Estimation under a possibly wrong error distribution; coefficients stay consistent, inference does not.
Leverage effect	Negative shocks raise volatility more than positive shocks of equal size.
Engle-Ng sign-bias test	Regresses squared residuals on the sign and size of the lagged shock to test for asymmetry.
EGARCH	Models $\log(\sigma^2)$ with separate size and sign terms; no non-negativity constraints needed.
GARCH-M	The conditional variance enters the conditional mean, capturing a risk premium.
Value-at-Risk (VaR)	A tail-based risk measure; requires the assumed error distribution to be correct, not just the standard errors.

Review questions

Use these to check your understanding before the seminar.

1. Explain how a series can be homoskedastic unconditionally but display volatility clustering. Which variance does each statement refer to?
2. Give reasons a forecaster or investor would want a forecast of the variance rather than just the mean.
3. Write down the ARCH(1) model. Why must α_0 be strictly positive, and why must α_1 be less than one?
4. For an ARCH(q), state the stationarity condition and the resulting unconditional variance. What happens if the condition fails?
5. Set out the five steps of the ARCH LM test. What exactly is the null hypothesis?
6. Why test at lag 1 first and then at lag 5 for daily data? What would each result tell you?
7. After fitting an ARCH(1) you fail to reject at lag 1 but reject at lag 5. What do you conclude and what do you do next?
8. The fitted conditional volatility rises sharply and falls slowly. Which feature of the model produces this asymmetry in shape, and what would a spiky series indicate instead?
9. Why does a pure ARCH need many lags to fit financial volatility? Explain what GARCH does differently.
10. In the Apple GARCH(1,1), $\alpha_1 = 0.0987$ and $\phi_1 = 0.8562$. Compute persistence, check stationarity, and interpret the relative size of the two coefficients.
11. Why is an R^2 of approximately zero in the mean equation not a problem here?
12. Non-normal residuals do not break the GARCH coefficient estimates. What exactly do they break, and why?
13. Compare the ARCH(5) and GARCH(1,1) volatility series. They identify the same crises – so on what do they disagree, and which is right?
14. Explain why a symmetric GARCH cannot capture the leverage effect. Point to the specific feature of the variance equation responsible.

15. Describe the leverage effect mechanism through the debt-to-equity ratio. Give one reason the label is broader than the mechanism.

16. In the Engle-Ng regression, why can you not include a constant, S^+ and S^- simultaneously? What does the constant then measure?

17. Write the EGARCH variance equation. Explain the role of each of the four terms.

18. Why must α_1 multiply the absolute value $|z_{t-1}|$ rather than z_{t-1}? What would happen to the model if the bars were dropped?

19. Given $\alpha_1 = 0.1725$ and $\gamma = -0.1428$, compute the impact on log-variance of a positive and a negative shock, and state the ratio.

20. Why is the EGARCH stationarity condition simply $\phi_1 < 1$, with no non-negativity restrictions?

References and further reading

Asteriou, D., Hall, S.G. Applied Econometrics – Chapter 14.